

Q1 2021 (01 Sep Shift 2)

The average translational kinetic energy of N_2 gas molecules at $^{\circ}C$ becomes equal to the K.E. of an electron accelerated from rest through a potential difference of 0.1 volt.

(Given $k_B = 1.38 \times 10^{-23} \text{ J/K}$)

(Fill the nearest integer).

Q2 2021 (31 Aug Shift 2)

A mixture of hydrogen and oxygen has volume 500 cm^3 , temperature 300 K , pressure 400 kPa and mass 0.76 g . The ratio of masses of oxygen to hydrogen will be :-

(1) 3 : 8

(2) 3 : 16

(3) 16 : 3

(4) 8 : 3

Q3 2021 (31 Aug Shift 1)

For an ideal gas the instantaneous change in pressure 'p' with volume 'v' is given by the equation $\frac{dp}{dv} = -ap$.

If $p = p_0$ at $v = 0$ is the given boundary condition, then the maximum temperature one mole of gas can attain is : (Here R is the gas constant)

(1) $\frac{p_0}{aeR}$

(2) $\frac{ap_0}{eR}$

(3) infinity

(4) $0^{\circ}C$

Q4 2021 (27 Aug Shift 2)

If the rms speed of oxygen molecules at $0^{\circ}C$ is 160 m/s , find the rms speed of hydrogen molecules at $0^{\circ}C$.

(1) 640 m/s

(2) 40 m/s

(3) 80 m/s

(4) 332 m/s

Q5 2021 (27 Aug Shift 1)

A balloon carries a total load of 185 kg at normal pressure and temperature of 27°C . What load will the balloon carry on rising to a height at which the barometric pressure is 45 cm of Hg and the temperature is -7°C . Assuming the volume constant?

(1) 181.46 kg

(2) 214.15 kg

(3) 219.07 kg

(4) 123.54 kg

Q6 2021 (26 Aug Shift 2)

A cylindrical container of volume $4.0 \times 10^{-3} \text{ m}^3$ contains one mole of hydrogen and two moles of carbon dioxide. Assume the temperature of the mixture is 400 K. The pressure of the mixture of gases is :

[Take gas constant as $8.3 \text{ J mol}^{-1} \text{ K}^{-1}$]

(1) $249 \times 10^1 \text{ Pa}$

(2) $24.9 \times 10^3 \text{ Pa}$

(3) $24.9 \times 10^5 \text{ Pa}$

(4) 24.9 Pa

Q7 2021 (26 Aug Shift 1)

The rms speeds of the molecules of Hydrogen, Oxygen and Carbon dioxide at the same temperature are

V_H , V_O and V_C respectively then :

(1) $V_H > V_O > V_C$

(2) $V_C > V_O > V_H$

(3) $V_H = V_O > V_C$

(4) $V_H = V_O = V_C$

Answer Key

Q1 (500)

Q2 (3)

Q3 (1)

Q4 (1)

Q5 (4)

Q6 (3)

Q7 (1)

Q1 (500)

Given

Translation K.E. of $N_2 = K.E.$ of electron

$$\frac{3}{2}kT = eV$$

$$\frac{3}{2} \times 1.38 \times 10^{-23} T = 1.6 \times 10^{-19} \times 0.1$$

$$\Rightarrow T = 773K$$

$$T = 773 - 273 = 500^\circ C$$

Q2 (3)

$$PV = nRT$$

$$400 \times 10^3 \times 500 \times 10^{-6} = n \left(\frac{25}{3} \right) (300)$$

$$n = \frac{2}{25}$$

$$n = n_1 + n_2$$

$$\frac{2}{25} = \frac{M_1}{2} + \frac{M_2}{32}$$

Also $M_1 + M_2 = 0.76gm$

$$\frac{M_2}{M_1} = \frac{16}{3}$$

Q3 (1)

$$\int_{p_0}^p \frac{dp}{P} = -a \int_0^v dv$$

$$\ln \left(\frac{p}{p_0} \right) = -av$$

$$p = p_0 e^{-av}$$

For temperature maximum $p - v$ product should be maximum

$$T = \frac{pv}{nR} = \frac{p_0 v e^{-av}}{R}$$

$$\frac{dT}{dv} = 0 \Rightarrow \frac{p_0}{R} \{e^{-av} + v e^{-av}(-a)\}$$

$$\frac{p_0 e^{-av}}{R} \{1 - av\} = 0$$

$$v = \frac{1}{a}, \infty$$

$$T = \frac{p_0^1}{Rae} = \frac{p_0}{Rae}$$

$$\text{at } v = \infty$$

$$T = 0$$

Option (1)

Q4 (1)

$$V_{\text{rms}} = \sqrt{\frac{3KT}{M}}$$

$$\frac{(V_{\text{rms}})_{\text{O}_2}}{(V_{\text{rms}})_{\text{H}_2}} = \sqrt{\frac{M_{\text{H}_2}}{M_{\text{O}_2}}} = \sqrt{\frac{2}{32}}$$

$$(V_{\text{rms}})_{\text{H}_2} = 4 \times (V_{\text{rms}})_{\text{O}_2}$$

$$= 4 \times 160$$

$$= 640 \text{ m/s}$$

Q5 (4)

$$P_m = \rho RT$$

$$\therefore \frac{P_1}{P_2} = \frac{\rho_1 T_1}{\rho_2 T_2}$$

$$\frac{\rho_1}{\rho_2} \Rightarrow \frac{P_1 T_2}{P_2 T_1} = \left(\frac{76}{45}\right) \times \frac{266}{300}$$

$$\frac{\rho_1}{\rho_2} \Rightarrow \frac{M_1}{M_2} = \frac{76 \times 266}{45 \times 300}$$

$$\therefore M_2 \Rightarrow \frac{45 \times 300 \times 185}{76 \times 266} = 123.54 \text{ kg}$$

Q6 (3)

$$V = 4 \times 10^{-3} \text{ m}^3$$

$$n = 3 \text{ moles}$$

$$T = 400 \text{ K}$$

$$PV = nRT \Rightarrow P = \frac{nRT}{V}$$

$$P = \frac{3 \times 8.3 \times 400}{4 \times 10^{-3}}$$

$$= 24.9 \times 10^5 \text{ Pa}$$

Ans (3)

Q7 (1)

$$V_{\text{RMS}} = \sqrt{\frac{3RT}{M_w}}$$

$$\text{At the same temperature } V_{\text{RMS}} \propto \frac{1}{\sqrt{M_w}}$$

$$\Rightarrow V_H > V_O > V_C$$

Option (1)